A self-similar Lorentz counterexample: an optimized proof and exact certificate

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Abstract

We construct a continuous nonodd function on $[-1,1]$ whose best uniform polynomial approximant of every degree vanishes at the origin. A finite certificate handles degrees zero through six; cubic self-similarity handles all remaining degrees. Keeping positive and negative contact sets separate reduces the analytic persistence argument to one inequality and one smallness condition. The construction works for every $0 < \varepsilon \le 4 \cdot 10^{-8}$, including the earlier choice 10^{-10}. Exact finite checks are reproducible from the accompanying Lean coefficient tables.

1 Construction and reduction to seven degrees

Write $I = [-1,1]$, let $T_k(\cos\theta) = \cos(k\theta)$, and let $P_n f$ denote the unique best uniform approximant to $f \in C(I)$ among polynomials of degree at most n. We address the origin interpolation implication discussed in [1]. Set $z = 2\cos(\pi/36)$ and

$$\begin{aligned} 107\beta = {} & -60 + 82z + 1086z^2 + 1576z^3 - 2932z^4 - 3433z^5 \\ & + 2002z^6 + 2243z^7 - 510z^8 - 587z^9 + 44z^{10} + 54z^{11}. \end{aligned}$$

In particular $419/1000 < \beta < 420/1000$. Define

$$h = T_4 - 1 - 2T_5 + \beta T_7, \qquad q_\varepsilon = T_3 + \varepsilon h, \qquad f_\varepsilon = \sum_{j\ge 0} (-1/3)^j q_\varepsilon \circ T_{3^j}.$$

The series converges uniformly on I, so f_ε is continuous.

Theorem 1. *For every $0 < \varepsilon \le 1/25000000$, the above function satisfies*

$$(P_n f_\varepsilon)(0) = 0 \quad (n \ge 0), \qquad f_\varepsilon(1/\sqrt{2}) + f_\varepsilon(-1/\sqrt{2}) = -3\varepsilon.$$

Thus it is not odd.

The reduction to finitely many degrees uses the identity

$$P_n(g \circ T_M) = (P_{\lfloor n/M \rfloor} g) \circ T_M \qquad (M \ge 1). \tag{1}$$

To prove it, average a competitor $p(\cos\theta)$ over $\theta + 2\pi j/M$, $0 \le j < M$. The target is invariant under these shifts, and the average cannot increase its uniform error. Only cosine frequencies divisible by M remain, giving $r(\cos M\theta)$ with $\deg r \le \lfloor n/M \rfloor$. Surjectivity of $T_M : I \to I$ gives equality of the optimal errors, and uniqueness of best approximation gives (1).

The functional equation and its consequence for $n \ge 7$ are

$$f_\varepsilon = q_\varepsilon - \tfrac{1}{3} f_\varepsilon \circ T_3, \qquad P_n f_\varepsilon = q_\varepsilon - \tfrac{1}{3}(P_{\lfloor n/3 \rfloor} f_\varepsilon) \circ T_3. \tag{2}$$

Here we used polynomial translation, scalar homogeneity, and $\deg q_\varepsilon \le 7$. Since $q_\varepsilon(0) = T_3(0) = 0$, induction reduces the origin condition to $0 \le n \le 6$.

2 Baseline residuals and the finite certificate

Put

$$a = 1/\sqrt{2}, \quad G = \sum_{j\geq 0}(-1/3)^j T_{3^j}, \quad F = G \circ T_3, \quad H = \sum_{j\geq 0}(-1/3)^j h \circ T_{3^j}.$$

Then $f_\varepsilon = F + \varepsilon H$. Pairing consecutive terms gives

$$G = \sum_{j\geq 0} 9^{-j} g \circ T_{9^j}, \qquad g(x) = 2x - \tfrac{4}{3}x^3.$$

The unique maximum of g on I is at a, its unique minimum is at $-a$, and $T_9(\pm a) = \pm a$. Hence G has the same unique extrema and

$$E := \|G\|_\infty = \frac{g(a)}{1 - 1/9} = \frac{3a}{2}.$$

Oddness gives $P_0 G = 0$. From $G = x - G \circ T_3/3$ and (1), $P_1 G = P_2 G = x$. Thus for the seven base degrees,

$$p_n^0 = P_n F = \begin{cases} 0 & n \leq 2, \\ T_3 & 3 \leq n \leq 6, \end{cases} \qquad b_n := F - p_n^0 = \begin{cases} G \circ T_3 & n \leq 2, \\ -G \circ T_9/3 & 3 \leq n \leq 6. \end{cases} \tag{3}$$

Their norms E_n are respectively E and $E/3$. Their contact sets S_n are respectively the zeros of T_6 and T_{18}.

Here are exact definitions of the finite certificate. Use the increasing nodes

$$X_j = -\cos\frac{(2j-1)\pi}{36} \quad (1 \leq j \leq 18), \qquad (C_1, \ldots, C_6) = (X_2, X_5, X_8, X_{11}, X_{14}, X_{17}).$$

The residual signs on the fine grid repeat $(+,-,-,+)$; call them σ_j. On the coarse grid they repeat $(-,+,+,-)$; call them τ_j. The two-cycle $T_3(a) = -a$, $T_3(-a) = a$ gives

$$H(\pm a) = -\tfrac{3}{2} \pm \tfrac{3}{2}(2+\beta)a.$$

Iterating $H = h - H \circ T_3/3$ twice, and using $T_{21} = -T_{15}$ at the zeros of T_{18}, yields $H(X_j) = R_\beta(X_j)$, where

$$R_\beta = T_4 - \tfrac{1}{3}T_{12} - \tfrac{5}{6} - 2T_5 + \beta T_7 + \tfrac{2+\beta}{3}T_{15} + \tfrac{2+\beta}{6}T_9. \tag{4}$$

Define corrections v_n and slopes λ_n as follows:

$$\begin{aligned} v_0 &= 0, & \lambda_0 &= \tfrac{\sqrt{6}}{4}(2+\beta), \\ v_1 = v_2 &= -(2+\beta)x, & \lambda_1 = \lambda_2 &= \tfrac{2+\beta}{2\sqrt{2}}, \\ v_3 &= A_1 x + A_2 x^2 + A_3 x^3, \\ v_4 &= T_4 - 1 + B_1 T_1 + B_3 T_3, \\ v_5 = v_6 &= T_4 - 1 + D_1 T_1 + D_3 T_3 + D_5 T_5, & \lambda_6 &= \lambda_5. \end{aligned}$$

The remaining coefficients and slopes are the solutions of

$$v_n(X_j) + \sigma_j \lambda_n = R_\beta(X_j), \tag{5}$$

n	j
3	$1, 3, 8, 10$
4	$10, 16, 18$
5	$10, 13, 15, 16.$

The three defining matrices are nonsingular, as certified below. In particular $\deg v_n \le n$ and $v_n(0) = 0$.

For a contact $\xi \in S_n$, write $s(\xi) = \operatorname{sgn} b_n(\xi)$. The exact finite calculation establishes

$$s(\xi)\big(H(\xi) - v_n(\xi)\big) \le \lambda_n, \qquad 0 < \lambda_n < 2, \tag{6}$$

with the following complete zero-slack sets. Indices refer to the indicated grid.

n	grid	zero-slack indices	other slacks exceed
0	C	$3, 4$	$127/50$
1,2	C	$1, 3, 4, 6$	$363/100$
3	X	$1, 3, 8, 10, 16$	$9/100$
4	X	$1, 3, 8, 9, 10, 16, 18$	$7/10$
5,6	X	$3, 4, 6, 9, 10, 13, 15, 16$	$1/5$

For $n = 0, 2, 3, 6$, use the whole listed set. For $n = 1, 4, 5$ use respectively $\{1, 3, 4\}$, $\{1, 3, 9, 10, 16, 18\}$, and $\{3, 4, 6, 9, 10, 13, 15\}$. Each selected set contains $n + 2$ increasing nodes with alternating signs.

The sums of the absolute Chebyshev coefficients of v_n are respectively

$$0, \quad < 5/2, \quad < 5/2, \quad < 9/4, \quad < 13/4, \quad < 11/2, \quad < 11/2.$$

Consequently $\|v_n\|_\infty < 6$ and $\|v_n'\|_\infty < 216$, since $\|T_k'\|_\infty = k^2$ and $\deg v_n \le 6$.

Exact verification of the finite assertions

All the numbers above lie in $\mathbb{Q}[z]$, with arithmetic reduced modulo

$$m(z) = z^{12} - 12z^{10} + 54z^8 - 112z^6 + 105z^4 - 36z^2 + 1 = (z^6 - 6z^4 + 9z^2 - 2)^2 - 3.$$

The last identity and $z = 2\cos(\pi/36)$ give $m(z) = 0$. There is a unique root in $(1.98, 2)$: the inner polynomial is positive there and has derivative $6z(z^2-1)(z^2-3) > 0$. Rational endpoint evaluations place the root in

$$[199238939/10^8,\ 199238940/10^8].$$

The accompanying `scripts/check_certificate.py` reads exact rational coefficient lists from `CertificateData.lean` and `CertificateGrid.lean`. It reconstructs the nodes by the Chebyshev recurrence, verifies (4), the functional equation on the grid, every slack identity, every equality set, alternation, coefficient norms, and the nonzero determinants in (5). Equality is tested as equality of reduced rational polynomials. One hundred rational bisections, followed by rational interval Horner evaluation, certify the strict inequalities. No floating-point tolerance is used. This also verifies that the stored coefficients are the uniquely defined solutions of (5).

3 Signed cusps and a global bound relative to contacts

The two estimates needed to turn (6) into uniform approximation are proved here with explicit constants.

Lemma 1. *For $s \in \{-1, 1\}$ set $S_{n,s} = \{\xi \in I : b_n(\xi) = sE_n\}$. For $0 \le n \le 6$ and $x \in I$,*

$$E_n - sb_n(x) \ge \frac{1}{250} \operatorname{dist}(x, S_{n,s}).$$

Proof. First we prove $E - sG(y) \geq c_G|y - sa|$ on I, where $c_G = \pi(\sqrt{2}-1)/324$. The factorization

$$g(a) - g(y) = \tfrac{4}{3}(y-a)^2(y+\sqrt{2}) \geq C_0(y-a)^2, \qquad C_0 = \tfrac{4}{3}(\sqrt{2}-1),$$

makes every term of $E - G(y) = \sum_{j\geq 0} 9^{-j}(g(a) - g(T_{9^j}(y)))$ nonnegative. Put $\delta = \pi/12$ and write $y = \cos(\pi/4 + u)$. For $0 < |u| \leq \delta$, choose the largest $j \geq 0$ with $9^j|u| \leq \delta$, so $\delta/9 < 9^j|u| \leq \delta$. Since $9^j \equiv 1 \pmod 8$ and $\sin t \geq 1/2$ for $t \in [\pi/6, \pi/3]$, this single summand gives

$$E - G(y) \geq \frac{C_0}{4} 9^{-j}(9^j u)^2 > \frac{C_0\delta}{36}|u| \geq c_G|y-a|.$$

For $|u| > \delta$, we have $|y - a| \geq r_0 = (\sqrt{3} - \sqrt{2})/2$. The first summand then gives the result because $C_0 r_0 > c_G$. At $u = 0$ it is immediate. Oddness of G proves the negative-sign case.

For every integer $L \geq 1$ and $s = \pm 1$ we also have the signed pullback bound

$$|T_L(x) - sa| \geq \kappa L \operatorname{dist}(x, \{\xi : T_L(\xi) = sa\}), \qquad \kappa = 4(1-a)/\pi. \tag{7}$$

To see this, on $[0, \pi]$ the inequality $|\cos t - a| \geq \kappa|t - \pi/4|$ follows from the concavity chord on $[0, \pi/4]$ and $\sin t \geq a > \kappa$ on $[\pi/4, \pi/2]$. On $[\pi/2, \pi]$, convexity gives $\cos t \leq 1 - 2t/\pi$, and

$$a - \cos t \geq 2t/\pi - (1-a) \geq 4(1-a)(t - \pi/4)/\pi,$$

because $a \geq 1/2$. Reflection gives the analogous assertion for $-a$. Apply these inequalities on the half-period of $L\theta$ containing the given angle, where $x = \cos\theta$. The unique preimage of the prescribed target in that half-period lies in $[0, \pi]$ after division by L. The inequality $|\cos\theta - \cos\phi| \leq |\theta - \phi|$ proves (7).

For $n \leq 2$ use $L = 3$ and target sa; for $n \geq 3$ use $L = 9$ and target $-sa$, with the factor $1/3$ in (3). Both give

$$E_n - sb_n(x) \geq 3c_G\kappa \operatorname{dist}(x, S_{n,s}), \qquad 3c_G\kappa = \frac{3 - 2\sqrt{2}}{27\sqrt{2}} > \frac{1}{250}.$$

The final strict inequality is equivalent to $750 > 527\sqrt{2}$, verified by squaring positive quantities. □

Lemma 2. *For $r_n = H - v_n$, every $0 \leq n \leq 6$, every contact $\xi \in S_n$, and every $x \in I$,*

$$|r_n(x) - r_n(\xi)| \leq 10^5|x - \xi|.$$

This is a bound relative to each fixed contact; it asserts no global Lipschitz regularity of H.

Proof. Pair the perturbation series:

$$H = \sum_{j\geq 0} 9^{-j} Q \circ T_{9^j}, \qquad Q = h - h \circ T_3/3.$$

Since $T_3'(\pm a) = 3$ and $h_{\text{even}} = T_4 - 1$, we have $Q'(\pm a) = 0$. For $x = \cos(\pi/4 + u)$, $0 < |u| \leq \delta = \pi/12$, choose the last scale j as in Lemma 1. Taylor's theorem and a geometric tail bound give

$$\begin{aligned} |H(x) - H(a)| &\leq \tfrac{1}{2}\left\|Q''\right\|_\infty u^2 \sum_{k=0}^{j} 9^k + 2\left\|Q\right\|_\infty \sum_{k=j+1}^{\infty} 9^{-k} \\ &\leq \left(\tfrac{9\delta}{16}\left\|Q''\right\|_\infty + \tfrac{9}{4\delta}\left\|Q\right\|_\infty\right)|u|. \end{aligned}$$

Here $|x-a| \geq |u|/2$. The same proof around $3\pi/4$ applies to $-a$. Thus, whenever $|x \mp a| \leq r_0$, a valid constant is

$$L_H = \tfrac{3\pi}{32} \left\|Q''\right\|_\infty + \tfrac{54}{\pi} \|Q\|_\infty < 7000.$$

For the numerical bound, $\beta < 0.420$ and the Chebyshev derivative formulas $\|T_k'\|_\infty = k^2$, $\|T_k''\|_\infty = k^2(k^2-1)/3$ give

$$\|h\|_\infty < 4.42, \qquad \left\|h'\right\|_\infty < 87, \qquad \left\|h''\right\|_\infty < 810,$$

and hence $\|Q\|_\infty < 5.894$ and $\|Q''\|_\infty \leq 28 \|h''\|_\infty + 8 \|h'\|_\infty < 23376$. Using $3 < \pi < 22/7$ yields the displayed bound for L_H.

If $L = 3^k$ and $T_L(\xi) = \pm a$, iterate the functional equation k times. The derivative bound for its finite prefix is $\|h'\|_\infty \sum_{i=0}^{k-1} 3^i = (L-1)\|h'\|_\infty/2$. Since $\|T_L'\|_\infty = L^2$, this gives

$$|H(x) - H(\xi)| \leq \left(\tfrac{L-1}{2} \left\|h'\right\|_\infty + L L_H\right) |x-\xi| \quad \text{if } |x-\xi| \leq r_0/L^2.$$

For our two grids $L = 3, 9$, and $r_0/81 > 10^{-3}$. Adding the derivative bound on v_n gives a constant less than $4 \cdot 87 + 9 \cdot 7000 + 216 < 10^5$ when $|x-\xi| \leq 10^{-3}$. The inequality is non-strict, including $x = \xi$.

Finally $\|H\|_\infty \leq \frac{3}{2}\|h\|_\infty < 6.63$, so $\|r_n\|_\infty < 20$. For $|x-\xi| > 10^{-3}$, $|r_n(x) - r_n(\xi)| < 40 < 10^5|x-\xi|$, which extends the estimate to all I. □

4 One persistence inequality and completion

Fix a base degree n and a sign $s \in \{-1, 1\}$. Choose a nearest contact $\xi \in S_{n,s}$ and put $d = |x-\xi|$. Lemmas 1 and 2, together with (6), give

$$\begin{aligned} s\big(b_n(x) + \varepsilon r_n(x)\big) &\leq E_n - d/250 + \varepsilon\big(\lambda_n + 10^5 d\big) \\ &= E_n + \varepsilon\lambda_n - (1/250 - 10^5\varepsilon)d \\ &\leq E_n + \varepsilon\lambda_n \qquad \left(0 \leq \varepsilon \leq \frac{1}{25000000}\right). \end{aligned}$$

Apply both signs. This proves

$$\left\|f_\varepsilon - (p_n^0 + \varepsilon v_n)\right\|_\infty \leq E_n + \varepsilon\lambda_n.$$

At the $n+2$ alternating equality nodes the residual is exactly $s(\xi)(E_n + \varepsilon\lambda_n)$. Its magnitude is positive, since $E_n > 0$ and $\lambda_n > 0$. The alternation theorem identifies $P_n f_\varepsilon = p_n^0 + \varepsilon v_n$. All these polynomials vanish at zero, and (2) proves the assertion for every degree.

For nonoddness, F is odd and T_3 interchanges a and $-a$. At both points $T_4 - 1 = -2$, so

$$H(a) + H(-a) = \sum_{j \geq 0} (-1/3)^j (-4) = -3.$$

It follows that $f_\varepsilon(a) + f_\varepsilon(-a) = -3\varepsilon \neq 0$, completing Theorem 1.

Reproduction and the formalization boundary

Run `python3 scripts/check_certificate.py` for the independent finite check, and `lake build` followed by `lake env lean Audit.lean` for the formal proof.

The shared module `PolynomialCertificate.lean` proves that exact rational coefficient certificates imply the real identities; the concrete checks use `decide +kernel`. The series pairing proof gives its summand explicitly to avoid expensive higher-order inference.

The Lean endpoint proves the unconditional existence of a sufficiently small positive parameter, with only `propext`, `Classical.choice`, and `Quot.sound`. The new generic signed persistence lemma is also checked in Lean. The explicit constants and the range $0 < \varepsilon \leq 4 \cdot 10^{-8}$ are proved in this document; they are not claimed as a quantitative Lean corollary. The original PDF is retained for comparison. Its strict Lipschitz inequality at coincident points is corrected here to $\leq$.

References

[1] E. B. Saff and R. S. Varga, *Remarks on some conjectures of G. G. Lorentz*, Journal of Approximation Theory **30** (1980), 29–36. https://www.math.kent.edu/~varga/pub/paper_126.pdf.